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Regulatory Pressure and Fire Sales in the Corporate Bond Market

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1 Big picture

- **Question.** Do regulatory capital constraints force insurance companies to sell downgraded corporate bonds, and do these sales create temporary fire-sale discounts in the corporate bond market?
- **Setting.** The empirical laboratory is the market for investment-grade corporate bonds that are downgraded to speculative grade, or “fallen angels,” during 2001–2005.
- **Why insurance companies matter.** Insurance companies hold a large share of investment-grade corporate bonds and face risk-based capital rules that become much more stringent when bonds fall below investment grade.
- **Key mechanism.** A downgrade changes the regulatory treatment of the bond. Constrained insurers then have incentives to sell, but other insurers face the same regulatory rule and cannot easily absorb the supply. Liquidity must therefore come from outside buyers such as high-yield bond funds and distressed hedge funds.
- **Main trading result.** Insurers with lower risk-based capital ratios are more likely to sell downgraded bonds immediately after the downgrade, and some selling begins even before the formal event.
- **Main price result.** Bonds with high predicted regulatory-induced selling experience large negative median cumulative abnormal returns around the downgrade and then reverse over subsequent weeks.
- **Magnitude.** The paper highlights a roughly 6–7 percentage point difference in MCARs between high- and low-regulatory-pressure bonds around the downgrade window.
- **Time variation.** Fire-sale effects are strongest when the insurance industry is stressed and when the capital of outside buyers is scarce.
- **Interpretive tension.** Credit downgrades contain information. The paper therefore distinguishes information effects from regulatory fire-sale effects using issuer stock returns, abnormal-return controls, and cross-sectional variation in insurers’ regulatory constraints.
- **Takeaway.** Regulation can create synchronized selling in illiquid markets. The same rule that limits risk-taking by insurers can amplify temporary price distortions when many constrained institutions hold the same asset.

2 Incremental contribution

- **New fire-sale channel.** Earlier fire-sale studies emphasize funding shocks, redemptions, aircraft liquidations, hedge-fund distress, or mutual-fund outflows. This paper identifies *regulatory pressure* as a distinct source of forced sales.
- **Institution-level identification.** The contribution is not just to show that downgraded bonds fall in price. The paper links individual insurers’ regulatory constraints to their actual trading decisions using NAIC transaction and position data.
- **Bond-level aggregation of seller constraints.** The paper moves from insurer-level constraints to bond-level fire-sale pressure by averaging the predicted selling probability or constraint exposure across the insurers holding each bond.
- **Separating forced selling from information.** Downgrades are informative events. The paper’s incremental contribution is to isolate the cross-sectional component of price pressure that is tied to constrained owners, not simply to news about default risk.
- **Slow-moving capital in corporate bonds.** The paper documents that prices recover only gradually, consistent with Duffie-style slow-moving capital in over-the-counter bond markets.
- **Sell-side and buy-side interaction.** It tests both sides of the Shleifer–Vishny mechanism: constrained sellers generate supply pressure, while scarce outside capital makes the price impact larger.

- **Regulation and market liquidity.** The paper shows that microprudential regulation can have market-level price effects when it pushes many regulated institutions in the same direction at the same time.
- **Policy relevance.** The evidence implies that capital rules can interact with illiquid markets and institutional asset holdings, creating temporary discounts that are not purely about fundamentals.

3 Data and measurement

3.1 Core data sources

- **NAIC transactions and holdings.** The paper uses insurer-level corporate bond transactions and positions from the National Association of Insurance Commissioners.
- **FISD bond characteristics.** Bond issue size, maturity, ratings, offering information, and related bond characteristics come from the Mergent Fixed Income Securities Database.
- **Street.com insurer data.** Insurance-company financial strength and risk-based capital information come from Street.com Ratings.
- **Market controls.** Stock returns, Treasury returns, investment-grade spread returns, high-yield spread returns, and issuer stock returns are used to control for fundamentals.
- **Potential buyer capital.** The paper measures outside buyer capacity using assets under management of high-yield mutual funds and high-yield/distressed hedge funds.

Interpretation: The data contribution is the merge of bond transactions, bond characteristics, insurer balance-sheet strength, and market prices. This allows the paper to connect regulation to trading and trading to prices.

3.2 Sample construction

- The raw NAIC transaction sample contains more than 1.7 million transactions over 2001–2005.
- The authors remove non-market trades, obvious data errors, non-dollar securities, very small issues, sinking-fund bonds, and bonds with problematic characteristics.
- The downgrade sample contains bonds downgraded from investment grade to speculative grade.
- The main downgrade sample has 1,179 downgraded bonds.
- The abnormal-return sample is smaller because reliable event-study tests require trades before, during, and after the downgrade window.

Interpretation: The sample screens are important because corporate bonds trade infrequently. The authors need enough trades to observe both the immediate price drop and later reversal.

3.3 Regulatory constraints

- The central insurer constraint measure is the NAIC risk-based capital ratio, or RBC ratio.
- Lower RBC ratios indicate less capital relative to regulatory risk charges and therefore tighter regulatory constraints.
- The paper also uses RARC1, a Street.com risk-adjusted capital ratio.
- Bonds are classified as high or low regulatory-pressure bonds using insurer-level predicted selling probabilities and the composition of the bond’s holders.

Interpretation: A downgrade increases the capital charge of a bond. The same downgrade is more painful for an insurer whose capital cushion is low.

3.4 Trading-volume variables

- Net volume equals insurer purchases minus insurer sales, usually scaled by bond, time window, or issue size.
- Negative net volume means insurers are net sellers.
- Event week zero is the downgrade announcement week.
- The paper studies several windows, including pre-downgrade weeks, the downgrade week, weeks 0 to +5, and later reversal windows.

Interpretation: The volume tests establish the mechanism: constrained insurers really do sell more, rather than prices merely falling for informational reasons.

3.5 Price impact and abnormal returns

- The main price outcome is the median cumulative abnormal return, or MCAR, around downgrade events.
- Abnormal returns adjust for bond fundamentals, Treasury returns, corporate spread returns, high-yield spread returns, aggregate stock returns, issuer stock returns, and transaction-cost effects.
- Price reversals are calculated by comparing fire-sale-window trade prices to later benchmark prices after recovery.
- Robustness checks use zero-issuer-stock-return bonds, mean adjustment, and bond-index adjustment.

Interpretation: The price tests focus on a temporary drop followed by reversal. This pattern is the distinguishing signature of fire-sale price pressure rather than a permanent information shock.

4 Models and empirical specifications

4.1 Probability of selling

The insurer-level probit model is:

$$\Pr(D_{i,j,k} = 1) = \Phi(\delta_0 + \delta_X X_{i,k} + \delta_Y Y_{i,j,k} + \delta_W W_k),$$

where $D_{i,j,k}$ indicates that insurer j sells bond i around downgrade event k , $X_{i,k}$ are bond characteristics, $Y_{i,j,k}$ are insurer and insurer-bond characteristics, and W_k are event controls.

Interpretation: A negative coefficient on $\log(RBC)$ or a positive coefficient on low-RBC dummies means that weaker, more constrained insurers sell more.

4.2 Trade price and abnormal return model

The observed price of bond i at trade date t is written as a fundamental value adjusted for transaction-cost direction:

$$P_{i,t} = (1 + A_{i,t} Q_{i,t}) E_t(V_i),$$

where $Q_{i,t} = 1$ for insurer buys and $Q_{i,t} = -1$ for insurer sells. The half-spread is modeled as:

$$A_{i,t} = \gamma_0 + \gamma_1 \log(S_{i,t}) + Z_{i,t}.$$

The logged trade-to-trade return is adjusted for market and issuer fundamentals:

$$\log(P_{i,t}/P_{i,t-1}) = \alpha + \beta Z_{i,t-1,t} + \text{transaction-cost controls} + \varepsilon_{i,t}.$$

Interpretation: This specification is designed to avoid confusing genuine bond-value news with transaction-price pressure.

4.3 MCAR event-study design

For each downgraded bond, abnormal returns are accumulated around event time and then aggregated across bonds using medians:

$$CAR_{i,t} = \sum_{\tau=-20}^t AR_{i,\tau}, \quad MCAR_t = \text{median}_i(CAR_{i,t}).$$

Interpretation: The paper emphasizes medians because corporate bond returns have outliers and skewness. Medians are more stable in this illiquid market setting.

4.4 Price reversal regression

The price reversal for trade t in bond i is:

$$R_{i,t,T} = \log(\bar{P}_{i,T}/P_{i,t}), \quad t \in [-5, +5], \quad T \geq +21.$$

The cross-sectional regression is:

$$R_{i,t,T} = \beta_0 + \beta_C C_i + \beta_X X_{i,t} + \beta_D D_i + \beta_Z Z_{t,T} + \beta_{ZD} D_i Z_{t,T} + \text{trade-cost controls} + \xi_{i,t,T}.$$

Interpretation: If bonds with more constrained owners have larger reversals after controlling for fundamentals, the original price drop is likely a temporary discount from forced selling.

4.5 Buyer-capital tests

$$\text{Price impact}_q = f(\text{insurance industry stress}_q, \text{potential buyers' capital}_q, \text{bond controls}).$$

Interpretation: Fire sales should be strongest when constrained sellers are numerous and specialized buyers have little capital. This is the sell-side/buy-side interaction in the fire-sale theory.

5 Identification and empirical design

5.1 Core identification challenge

A downgrade can lower bond prices because it reveals bad credit information, not because of forced selling. The paper must therefore separate three forces: fundamental news, ordinary price pressure, and regulatory-driven fire sales.

5.2 Main identifying variation

- **Across insurers:** some insurers have lower RBC ratios and therefore tighter regulatory constraints.
- **Across bonds:** some downgraded bonds are held disproportionately by constrained insurers.
- **Across time:** some quarters have higher insurance-industry distress and lower outside buyer capital.
- **Across event windows:** fire sales imply a drop followed by reversal; information implies a permanent repricing.

5.3 Why this design is persuasive

- The paper observes actual trades, not only holdings or prices.
- It predicts who should be forced to sell using pre-event balance-sheet constraints.
- It shows that the same constrained owners are more likely to sell.
- It shows that bonds with higher predicted forced selling have larger temporary price drops.
- It shows that fire-sale effects are larger when outside liquidity is scarce.
- It confirms the pattern in zero-issuer-stock-return bonds, where downgrade information should be limited.

5.4 What the design cannot fully prove

- Fundamental value is not directly observable.
- Bond liquidity and trade timing are irregular.
- Downgrades may be anticipated, so some pre-event price movement may combine information and forced-sale expectations.
- The paper measures regulatory constraints, but it cannot observe the exact internal decision rule of every insurer.
- Outside buyer capital is proxied by high-yield mutual fund and hedge fund AUM, not by the complete universe of buyers.

Interpretation: The empirical design is strong because several independent patterns line up: constrained insurers sell; their bonds fall more; prices reverse; effects are largest when buyers are scarce; and the pattern survives information controls.

6 Section-by-section study guide

- **Section 1.** Frames the paper as a test of regulatory-induced fire sales in corporate bonds.
- **Section 2.** Builds the transaction-level data set and defines downgraded bonds, insurer constraints, and descriptive statistics.
- **Section 3.** Shows that constrained insurers are more likely to sell fallen angels and that selling volume is concentrated around downgrades.
- **Section 4.** Shows that selling pressure maps into temporary price deviations and reversals.
- **Section 5.** Rules out alternative explanations including pure information effects, simple large-trade price pressure, and abnormal-return methodology choices.
- **Section 6.** Concludes that regulation-induced fire sales matter in corporate bond markets.

7 Tables and figures

7.1 Table 1: Data Screening and Merging Process + Table 2: Bond and Transaction Descriptive Information

Read:

- Table 1 starts from more than 1.7 million NAIC transactions and narrows the sample to 1,179 downgraded bonds.
- The screens eliminate non-market trades, outliers, nonstandard bonds, insufficient bond characteristic data, and bonds lacking adequate return observations.
- Table 2 shows that downgraded bonds are large, actively traded, and initially heavily held by insurers.
- Insurance holdings of downgraded bonds drop from roughly one-third of issue size before the downgrade to roughly one-fifth after the downgrade.

Detailed interpretation: Table 1 is the audit trail for the paper’s empirical sample. It shows that the final event-study sample is much smaller because corporate bonds are illiquid and many issues do not trade in all required windows. Table 2 establishes that the downgrade event coincides with a meaningful change in insurer participation, which makes the regulatory channel empirically plausible.

Economic message: The sample is not just a set of credit events; it is a set of events where institutional owners have a regulatory reason to trade and where trading is observable.

Table 1

Data screening and merging process.

This table describes how the data are screened and merged. The raw sample is the NAIC transaction data for the period from 2001 to 2005. First, the sample is cleaned and then merged with (i) the rating and FSD bond characteristics and (ii) the Street.com Ratings insurance company data. The downgrade sample is constructed by including only bonds that are downgraded from investment (BBB- and above) to non-investment (BB+ and below) grades during the sample period. The downgrade sample is merged with CRSP’s issuer stock data, and the bonds that exhibit insignificant stock return upon the downgrade are identified. The table reports the numbers of transactions, numbers of bond issues, numbers of issuers, and numbers of insurance companies behind the transactions.

	Transactions		Bond issues	Issuers (CUSIP)	Insurance companies
	Life & health	Property			
Total sample					
(1) Raw sample (2001–2005)	1,136,515	571,733	1,708,248	119,790	29,468
Data errors, non-secondary market trades, and potential outliers ^a	(630,101)	(258,860)	(888,961)	(57,323)	(12,336)
(2) Merged with credit ratings and FSD bond characteristics	399,470	264,876	664,346	23,276	6,040
Bonds that (i) are not debentures or medium-term notes (MTN), (ii) have issue size smaller than \$50 million, (iii) have sinking funds, or (iv) are denominated in foreign currencies. Trades that are before offering or after maturity	(111,144)	(65,888)	(177,032)	(9,186)	(1,812)
(3) Merged with the Street.com data ^b	288,180	195,530	483,710	14,074	4,228
Downgrade sample					
(4) Cleaned downgrade sample from merged data (3)	40,961	25,060	66,021	1,179	417
(5) Merged with CRSP stock information ^c	34,038	22,017	56,055	1,043	335
(5.1) Meeting requirements for abnormal return study ^d	19,692	14,030	33,722	356	173
(6) Exhibiting zero-issuer-stock returns upon downgrade ^e	20,250	13,133	33,383	584	221
(6.1) Meeting requirements for abnormal return study ^f	12,627	9,030	21,657	237	117

^a Examples of non-market counterparties include called, cancelled, conversion, direct, maturity, put, redemption, tax-free exchange, etc. An outlier is defined as a transaction whose product of immediately preceding and subsequent returns is negative and greater than 0.09 in absolute value.

^b The Street.com may not evaluate some insurance companies for any of the following reasons: (i) total assets < \$1 million, (ii) premium income for the current year is < \$100,000, (iii) the company functions almost exclusively as a holding company, or (iv) the Street.com does not have enough information.

^c Each bond is first matched with the issuer stock identifier through the six-digit issuer CUSIP. The unmatched bonds are then matched with the stocks through stock tickers of the bond issuers obtained from Bloomberg and TRACE.

^d For reliability, abnormal return studies use only bonds that trade: (i) at least once before week –20, (ii) at least once between weeks –20 and 0, (iii) at least once between weeks 0 and +35, (iv) at least once after week +35, and (v) at least ten times during the sample period.

^e Following Ambrose, Cai, and Helwege (2009), the zero-issuer-stock return sample is identified by comparing the average daily stock return over the three-

Table 2
Bond and transaction descriptive information.

This table provides descriptive information regarding the samples used in the empirical analyses. The sample period is from 2001 to 2005. The total sample includes all transactions and underlying bonds that pass the screening process in Table 1 up to step (3). The “downgrade” sample includes only transactions on bonds that are downgraded from investment (BBB- and above) to non-investment (BB+ and below) grades during the sample period. Average issue size is calculated across bond issues, taking each issue as one observation. For each issue, insurance holding is the (quarterly time-series) median holding of all insurance companies together, measured as a percentage of the bond’s issue size. Average holding is then calculated across bond issues, taking each issue as one observation. Buy trades refer to transactions in which the reporting insurance companies buy (mostly from dealers). For transaction characteristics, the averages and medians are calculated across transactions, taking each transaction as one observation. The holding and transaction statistics for the “downgrade” sample are calculated separately for the periods before and after the downgrade.

	Always investment-grade	Downgrade		Always non-investment-grade	Not rated	Total
		Before downgrade	After downgrade			
Number of issues	8,081	1,179		4,634	180	14,074
Average issue size (\$ million)	381.15	416.88	21.22	277.17	303.71	348.92
Average insurance holding (% of issue size)	34.10	34.09		8.47	11.63	25.05
Number of trades	311,699	42,181	23,840	102,503	3,487	487,170
Number of trades per issue	39	36	20	22	19	34
% Buy trades	59.95	52.67	41.67	49.16	41.76	56.00
Average trade characteristics						
Price (per \$100 par)	105.38	99.16	90.32	96.01	80.49	101.93
Size (\$ million)	3.22	3.11	2.32	1.55	1.97	2.80
Age (years)	2.75	2.36	4.20	2.22	2.24	2.67
Maturity (years)	9.56	10.63	9.73	7.28	6.58	9.16
Median trade characteristics						
Price (per \$100 par)	104.10	100.45	96.25	100.96	94.23	102.70
Size (\$ million)	1.20	1.00	0.85	0.50	0.71	1.00
Age (years)	1.85	1.70	3.64	1.56	1.73	1.86
Maturity (years)	7.42	7.84	6.52	6.97	6.47	7.22

(a) Table 1: Data screening and merging process

(b) Table 2: Bond and transaction descriptive information

7.2 Table 3: Insurance Company Descriptive Information + Table 4: Probability of Selling Around the Downgrade

Read:

- Table 3 describes insurer size, capital, RBC ratios, RARC1 ratios, and risky-asset holdings.
- Life and property insurers differ in size and balance-sheet structure, motivating separate specifications.
- Table 4 shows that lower RBC ratios predict higher probability of selling downgraded bonds.
- Property insurers have a higher selling probability than life insurers in the pooled regressions.
- Bonds downgraded below BB+ and older/smaller/more illiquid bonds also have different selling patterns.

Detailed interpretation: Table 4 is the first main causal-mechanism table. It moves the paper beyond price patterns and demonstrates that measured regulatory constraints predict actual sales. The marginal effects in Panel B convert probit estimates into economically interpretable changes in sale probabilities.

Economic message: Regulation affects actual trading decisions, not merely portfolio labels or accounting categories.

Table 3

Insurance company descriptive information.

This table provides descriptive information about the insurance companies that trade the bonds in the total sample and the “downgrade” sample. The sample period is from 2001 to 2005. The total sample includes all insurance companies behind the transactions that pass the screening process in Table 1 up to step (3). The “downgrade” sample includes only insurance companies behind the transactions on bonds that are downgraded from investment (BBB– and above) to non-investment (BB+ and below) grades during the sample period. The total number of trades only includes the trades that pass the sample screen in Table 1. All other variables are from the Street.com. The frequency is annual, and the variable values are for each year-end. The statistics reported are cross-sectional statistics based on the median value of each variable from 2000 to 2005. The detailed description of each variable is in Appendix A.

	Total sample										Downgrade sample									
	Life/health insurance (N=982)					Property/casualty insurance (N=2,060)					All insurance (N=2,451)									
	Mean	10th Pct	50th Pct	90th Pct		Mean	10th Pct	50th Pct	90th Pct		Mean	10th Pct	50th Pct	90th Pct		Mean	10th Pct	50th Pct	90th Pct	
<i>Measures of size</i>																				
Total number of bond trades (2001–2005)	293	5	73	795	95	3	34	218	195	10	60	473								
Invested assets (\$ million)	2,536	7	128	4,818	448	7	48	755	1,367	9	94	2,018								
Capital and surplus (\$ million)	270	3	31	589	187	4	24	303	251	4	35	463								
<i>Measures of regulatory constraint</i>																				
NAIC risk-based capital ratio (RBC ratio)	25.63	4.21	8.28	40.19	39.75	3.00	7.40	70.20	22.30	3.31	7.30	33.26								
Risk-adjusted capital ratio 1 (RACR1)	2.28	0.86	1.93	3.92	8.29	0.59	1.80	9.80	4.21	0.67	1.80	4.90								
<i>Measures of risk appetite</i>																				
Holding of non-investment-grade bonds (% of invested assets)	2.66	0.00	1.63	6.65	0.72	0.00	0.00	2.26	1.60	0.00	0.40	5.10								
Holding of risky assets (% of invested assets)	15.56	0.35	11.41	34.87	14.99	0.00	9.43	37.82	16.06	0.13	11.69	37.57								

(a) Table 3: Insurance company descriptive information (b) Table 4: Probability of selling around the downgrade

Table 4

Probability of selling around the downgrade.

This table reports probit estimates for the effects of regulatory constraints on the probability that an insurance company will sell the downgraded bonds during (i) event weeks 0 to +5 and (ii) event weeks –10 to 0. The sample period is from 2001 to 2005. The downgrade is from investment to non-investment grades. The dependent variable is a dummy that equals one if the insurance company (holding the bond) sells the downgraded bond during the period of interest, and zero otherwise. Panel A reports the coefficient estimates. Panel B reports the corresponding marginal effects evaluated at the sample means of the explanatory variables. In columns Life vs. property (1) and Life vs. property (2), the difference in selling propensity between life and property companies is analyzed. In columns Life (1) to Life (3), only life companies are used in the estimation. In columns Property (1) to Property (3), only property companies are used in the estimation. All models include state and year-quarter dummies. White's robust standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively. Variable definitions are in Appendices A and B.

	Selling during weeks 0 to +5					Selling during weeks –10 to 0		
	Life vs. property (1)	Life (1)	Life (2)	Property (1)	Property (2)	Life vs. property (2)	Life (3)	Property (3)
Panel A: Coefficient estimates								
Regulatory constraints								
Property insurance dummy	0.183*** (0.028)					0.232*** (0.026)		
ln(RBC ratio)		–0.262*** (0.060)		–0.202*** (0.070)			–0.129** (0.056)	–0.115* (0.062)
Dummy: RBC ratio ≤ 25th pct			0.146*** (0.048)		0.204*** (0.062)			
Dummy: 25th pct < RBC ratio ≤ 50th pct			0.184*** (0.044)		0.185** (0.072)			
Dummy: 50th pct < RBC ratio ≤ 75th pct			0.091** (0.045)		0.119* (0.071)			
Insurance company controls								
ln(Capital & surplus)		–0.009 (0.009)	–0.012 (0.009)	0.007 (0.014)	0.007 (0.014)		–0.009 (0.009)	0.014 (0.013)
Risky asset holding (% of portfolio)		0.001 (0.002)	0.001 (0.002)	–0.007*** (0.002)	–0.007*** (0.002)		–0.002 (0.002)	–0.005*** (0.001)
<i>Bond controls</i>								
Lower than BB+ dummy	0.090*** (0.026)	0.070** (0.032)	0.070** (0.032)	0.178*** (0.048)	0.178*** (0.048)	0.138*** (0.024)	0.152*** (0.031)	0.169*** (0.043)
ln(Age in years)	–0.120*** (0.014)	–0.123*** (0.018)	–0.124*** (0.018)	–0.111*** (0.026)	–0.110*** (0.026)	–0.108*** (0.013)	–0.132*** (0.017)	–0.061** (0.024)
ln(Issue size)	0.081*** (0.016)	0.076*** (0.020)	0.075*** (0.020)	0.076*** (0.029)	0.076*** (0.029)	0.152*** (0.015)	0.139*** (0.019)	0.159*** (0.025)
Observations	23,690	16,007	16,007	6,263	6,263	24,180	15,906	6,540
Pseudo R-squared	0.054	0.069	0.069	0.069	0.070	0.082	0.098	0.080
Log likelihood	–6,373	–4,081	–4,081	–2,043	–2,042	–7,584	–4,726	–2,546
Panel B: Marginal effects								
Regulatory constraints								
Property insurance dummy	0.026*** (0.004)					0.040*** (0.005)		
ln(RBC ratio)		–0.031*** (0.007)		–0.034*** (0.012)			–0.019** (0.008)	–0.024* (0.013)
Dummy: RBC ratio ≤ 25th pct			0.019*** (0.007)		0.035*** (0.011)			
Dummy: 25th pct < RBC ratio ≤ 50th pct			0.023*** (0.006)		0.034** (0.014)			
Dummy: 50th pct < RBC ratio ≤ 75th pct			0.011* (0.006)		0.021 (0.013)			
Insurance company controls								
ln(Capital & surplus)		–0.001 (0.001)	–0.001 (0.001)	0.001 (0.002)	0.001 (0.002)		–0.001 (0.001)	0.003 (0.003)
Risky asset holding (% of portfolio)		0.000 (0.000)	0.000 (0.000)	–0.001*** (0.000)	–0.001*** (0.000)		–0.000 (0.000)	–0.001*** (0.000)
<i>Bond controls</i>								
Lower than BB+ dummy	0.012*** (0.004)	0.008** (0.004)	0.008** (0.004)	0.031*** (0.008)	0.030*** (0.008)	0.023*** (0.004)	0.022*** (0.005)	0.016*** (0.009)
ln(Age in years)	–0.016*** (0.002)	–0.015*** (0.002)	–0.015*** (0.002)	–0.019*** (0.004)	–0.019*** (0.004)	–0.017*** (0.002)	–0.019*** (0.002)	–0.013** (0.005)
ln(Issue size)	0.011*** (0.002)	0.009*** (0.002)	0.009*** (0.002)	0.013*** (0.005)	0.013*** (0.005)	0.024*** (0.002)	0.020*** (0.003)	0.012*** (0.005)

7.3 Table 5: Insurance Companies’ Trading Volume Around the Downgrade + Figure 1: Net Volume by RBC Ratio

Read:

- Table 5 shows net selling spikes around the downgrade week and is especially strong for low-RBC insurers.
- The net volume is negative before and after the downgrade, indicating that selling pressure begins before the formal event and persists afterward.
- Figure 1 visualizes the same pattern: below-median RBC insurers sell much more than above-median RBC insurers.
- The largest selling occurs around week 0, where downgraded bonds enter the non-investment-grade category.

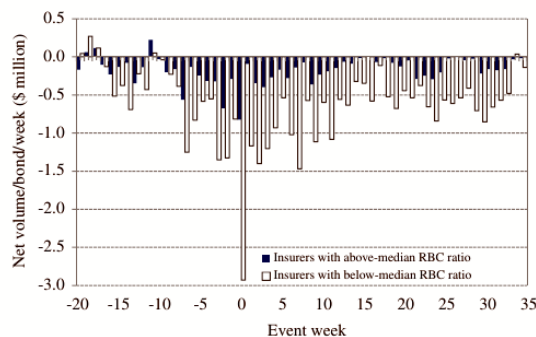
Detailed interpretation: This pair establishes timing. The regulatory rule binds most directly at the downgrade, but selling pressure is not confined to a single day. This is consistent with anticipatory selling, liquidity constraints, staggered insurer reactions, and slow trading in OTC corporate bond markets.

Economic message: The mechanism has the expected direction and timing: regulatory pressure produces coordinated net selling by constrained insurers.

Table 5
Insurance companies' trading volume around the downgrade.

This table reports insurance companies' trading volume of the downgraded bonds. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. A total of 1,179 downgraded bonds (sample 4 in Table 1) are considered. Week 0 is the downgrade announcement week. For each bond in each event week, the total volume is calculated as the sum of the buy volume and the sell volume, while the net volume is calculated as the buy volume minus the sell volume. In columns "All insurance," all insurance companies' trades are included while in columns "Below median RBC ratio," the volumes reflect only trades by insurance companies with RBC ratio below the sample median. Panel A reports the average total and net volumes across all bond-weeks during the defined period. The last column reports the p-values from one-sided sign tests of the null hypothesis that the number of bonds that are net sold is less than or equal to the number of bonds that are net bought. Panel B compares the average net volume between two groups of bonds with the average selling probability above and below the sample median. For each insurance company, the selling probability is calculated based on the model's Life (1) and Property (1) in Table 4. The bond-level average selling probability is calculated by averaging the selling probability across all insurance companies holding the bond at the end of the quarter preceding the downgrade.

5-Week period	Average total volume (\$ million/bond/week)		Below median RBC ratio as % of total	Net volume (\$ million/bond/week)		Net sell frequency < net buy frequency? Sign test (p-value)
	All insurance	Below median RBC ratio		All insurance	Below median RBC ratio	
Panel A: Net volume by event period						
[-30, -26]	1.019	0.830	72.5	0.020	-0.008	(0.991)
[-25, -21]	0.901	0.730	75.9	0.025	0.025	(0.835)
[-20, -16]	0.925	0.736	76.8	-0.061	-0.041	(0.002)
[-15, -11]	0.951	0.784	75.4	-0.180	-0.149	(0.000)
[-10, -6]	0.959	0.793	75.7	-0.307	-0.225	(0.000)
[-5, -1]	0.993	0.772	73.1	-0.433	-0.299	(0.000)
Week 0	2.482	1.976	73.8	-1.344	-1.058	(0.000)
[1, 5]	1.062	0.834	74.4	-0.492	-0.408	(0.000)
[6, 10]	0.883	0.732	78.2	-0.460	-0.382	(0.000)
[11, 15]	0.646	0.542	79.1	-0.289	-0.248	(0.000)
[16, 20]	0.597	0.475	76.9	-0.229	-0.199	(0.000)
[21, 25]	0.549	0.404	74.1	-0.315	-0.228	(0.000)
[26, 30]	0.509	0.420	79.4	-0.236	-0.213	(0.000)
[31, 35]	0.499	0.369	75.4	-0.186	-0.143	(0.000)
5-Week period	Above-median average selling probability			Below-median average selling probability		
	All insurance	Below median RBC ratio	Below median RBC ratio as % of total	All insurance	Below median RBC ratio	Below median RBC ratio as % of total
Panel B: Net volume for bonds with average selling probability above and below median						
[-10, -6]	-0.577	-0.401	73.5	-0.189	-0.148	77.8
[-5, -1]	-1.184	-0.875	71.5	-0.110	-0.052	75.0
Week 0	-3.201	-2.477	73.3	-0.553	-0.453	74.1
[1, 5]	-1.044	-0.814	72.7	-0.258	-0.235	75.9
[6, 10]	-0.896	-0.715	77.4	-0.274	-0.240	78.8



. This figure plots insurance companies' net trading volume of bonds that are downgr

(a) Table 5: Insurance companies' trading volume around the downgrade

(b) Figure 1: Net volume by RBC ratio

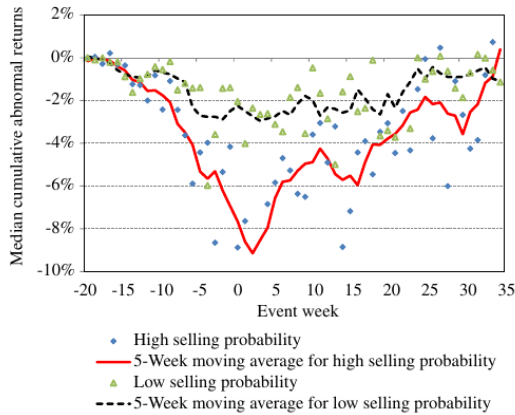
7.4 Figure 2: MCARs by Average Selling Probability + Table 6: Cumulative Abnormal Returns by Average Selling Probability

Read:

- Figure 2 shows that bonds with high predicted selling probability experience much larger price declines than low-selling-probability bonds.
- The high-pressure group bottoms out near the downgrade window and then recovers gradually.
- Table 6 quantifies the difference: MCARs are much more negative for the high-selling-probability group during week 0 and weeks 1–5.
- The difference largely fades by weeks 31–35, consistent with temporary price pressure rather than permanent information.

Detailed interpretation: This is the core price-evidence block. If downgrades simply revealed bad news, high- and low-regulatory-pressure bonds should not systematically differ after controlling for fundamentals. The strong decline and subsequent reversal for the high-pressure group is the fire-sale signature.

Economic message: Regulatory-induced selling creates economically large temporary discounts in corporate bonds.



(a) Figure 2: Median cumulative abnormal returns by average selling probability

Table 6
Cumulative abnormal returns of downgraded bonds by average selling probability.
This table reports the median cumulative abnormal returns (MCARs) for the downgraded bonds grouped by the average probability of selling during weeks 0 and ± 5 after the downgrade. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. A total of 1,179 downgraded bonds (sample 4 in Table 1) are considered. Week 0 is the downgrade announcement week. For each insurance company, the selling probability is calculated based on the models Life (1) and Property (1) in Table 4. The bond-level average selling probability is calculated by averaging the selling probability across all insurance companies holding the bond at the end of the quarter preceding the downgrade. The abnormal return for a bond from $t-1$ to t is measured as the residual ϵ_t from the model:

$$\ln(P_{t,T}/P_{t-1,T}) = \alpha + \beta Z_{t-1} + \gamma_0(Q_{t-1} - Q_{t-2}) + \gamma_1(Q_{t-1} \ln S_{t-1}) - Q_{t-1} \ln S_{t-1} + \beta + \epsilon_{t,T-1},$$

where Z is a vector of information variables that impact the bond's fundamental value, including duration-adjusted return on 10-year Treasury note, duration-adjusted return on Barclays investment-grade corporate bond spread, duration-adjusted return on Barclays high-yield corporate bond spread, CDS value-weighted return, and bond issuer's stock return. Q is an indicator variable equal to one if the transaction is a buy and -1 if it is a sell. S denotes trade size (par value). The parts associated with Q are included to control for the normal price impact of a trade. The model is estimated using the data for the period before week -20 to obtain the abnormal return from weeks -20 to -1 . The estimation is done separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and pre-downgrade rating at BBB $+$, (ii) maturity at downgrade > 5 years and pre-downgrade rating at BBB $+$, (iii) maturity at downgrade ≤ 5 years and pre-downgrade rating above BBB $+$, and (iv) maturity at downgrade > 5 years and pre-downgrade rating above BBB $+$. The model is then re-estimated using the data starting in week $+36$ to obtain the abnormal return from weeks 0 to $+35$. The estimation is done separately for four groups of bonds: (i) maturity at downgrade ≤ 5 years and post-downgrade rating at BB $+$, (ii) maturity at downgrade > 5 years and post-downgrade rating at BB $+$, (iii) maturity at downgrade ≤ 5 years and post-downgrade rating below BB $+$, and (iv) maturity at downgrade > 5 years and post-downgrade rating below BB $+$. The abnormal returns are then accumulated over time. The cumulative abnormal return for each bond is then normalized to zero in week -20 . MCARs are measured as across-bond across-transaction medians of cumulative abnormal returns for each five-week period relative to the event. Only 356 downgraded bonds in Sample 5.1 in Table 1 are used. The t -statistics are for the tests of (i) null hypothesis that MCAR equals zero and (ii) null hypothesis that MCARs for the two average selling probability groups are equal, and are calculated using bootstrapped standard errors (through median regressions of MCARs on a constant and a dummy for bonds with average selling probability above the median).

5-Week period	Above-median average selling probability		Below-median average selling probability		Difference	
	(1) MCAR	HO: MCAR=0 t-statistic	(2) MCAR	HO: MCAR=0 t-statistic	(1)-(2) MCAR	HO: (1)=(2) t-statistic
[-20, -16]	0.000	(0.00)	0.000	(0.00)	0.000	(0.00)
[-15, -11]	-0.814	(-4.07)	-0.879	(-4.25)	0.065	(0.16)
[-10, -6]	-2.987	(-6.18)	-0.967	(-3.38)	-2.020	(-3.31)
[-5, -1]	-5.323	(-6.77)	-2.147	(-3.65)	-3.176	(-3.11)
Week 0	-8.877	(-5.00)	-2.045	(-2.62)	-6.832	(-3.87)
[1, 5]	-8.680	(-9.22)	-2.824	(-4.49)	-5.857	(-5.27)
[6, 10]	-5.344	(-9.01)	-2.334	(-3.32)	-3.009	(-3.37)
[11, 15]	-4.276	(-5.35)	-2.430	(-3.83)	-1.846	(-1.72)
[16, 20]	-3.618	(-5.12)	-2.498	(-4.08)	-1.120	(-1.61)
[21, 25]	-2.233	(-3.42)	-2.430	(-3.23)	0.198	(0.22)
[26, 30]	-3.880	(-4.73)	-1.063	(-1.48)	-2.817	(-2.64)
[31, 35]	-1.101	(-0.75)	-0.527	(-0.99)	-0.574	(-0.14)

(b) Table 6: Cumulative abnormal returns by average selling probability

7.5 Table 7: Net Trading Volume and Impairment of Insurance Companies + Figure 3: Net Volume and Insurance Industry Impairment

Read:

- Table 7 relates selling to the insurance industry's impairment percentage.
- Net volume is most negative in years when insurers are under stress, especially 2001–2002 and 2005.
- Figure 3 plots the time series: high impairment periods align with greater net selling around downgrades.
- The pattern supports the idea that seller balance-sheet constraints are state dependent.

Detailed interpretation: This pair adds time-series evidence to the cross-sectional insurer-level evidence. If the same regulatory shock hits when the whole insurance sector is weak, many insurers try to sell together and few regulated peers can absorb the bonds.

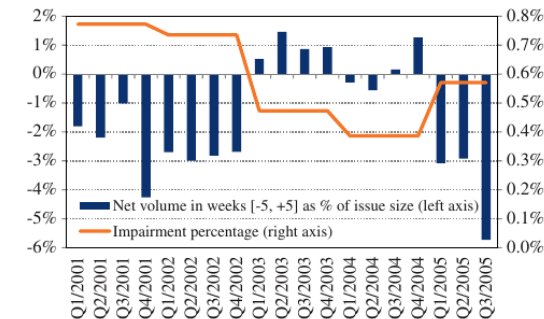
Economic message: Fire sales are not only about a bond's downgrade; they depend on the aggregate health of the institutions holding the bond.

Table 7
Net trading volume and impairment of insurance companies.

This table relates insurance companies' net trading volume of the downgraded bonds to the impairment ratio of the insurance industry. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. Impairment ratio is the number of insurance companies becoming impaired in a given year measured as a percentage of the total number of insurance companies. Impairment ratio is reported both for each type of insurer and as an asset-weighted average of the overall industry. For each bond, the net volume is calculated as the buy volume minus the sell volume during event weeks -5 and $+5$, reported as a proportion of the bond's issue size. In the last three columns, the percentage net volume is regressed on year dummies, bond characteristic controls, and the bond-level average selling probability, respectively (based on specifications Life (1) and Property (1) in Table 4). Bond characteristic controls include logged bond age at the time of downgrade, logged issue size, logged bond maturity at the time of downgrade, and the lower than BB $+$ dummy. All variables, other than the year dummies, are demeaned. Robust standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively.

Downgrade year	(1) Life & health impairment %	(2) Property & casualty impairment %	(3) Asset-weighted impairment %	(4) Net volume as proportion of issue size	(5) Net volume as proportion of issue size	(6) Net volume as proportion of issue size
2001	0.45	1.60	0.77	-0.0274*** (0.0035)	-0.0270*** (0.0035)	-0.0245*** (0.0034)
2002	0.45	1.50	0.74	-0.0283*** (0.0029)	-0.0296*** (0.0031)	-0.0290*** (0.0031)
2003	0.20	1.15	0.47	0.0106*** (0.0035)	0.0093*** (0.0038)	0.0041 (0.0047)
2004	0.30	0.60	0.39	0.0046 (0.0048)	0.0025 (0.0050)	-0.0053 (0.0067)
2005	0.60	0.50	0.57	-0.0309*** (0.0036)	-0.0343*** (0.0047)	-0.0307*** (0.0051)
Average selling probability						-0.1365** (0.0687)
Controlled for bond characteristics?				No	Yes	Yes
Observations	5	5	5	795	795	795
Adjusted R-squared				0.083	0.087	0.091

(a) Table 7: Net trading volume and impairment of insurance companies



industry's impairment percentage. This figure plots the time-series of insurance industry's impairment percentage (left axis). Impairment percentage is the number of insurance companies impaired divided by the total number of insurance companies. Plotted is the asset value-weighted average

(b) Figure 3: Net volume and insurance industry's impairment percentage

7.6 Table 8: Price Impact and Potential Buyers' Capital + Figure 4: Price Impact and Potential Buyers' Capital

Read:

- Table 8 shows that price impact is larger when potential buyer capital is low.
- The estimated price impact per million of insurer selling falls over time as high-yield bond fund and hedge-fund capital rises.
- Figure 4 visualizes the negative relationship between buyer capital and price impact.
- The evidence is strongest in 2001–2002, when buyers' capital is scarce and price impact is high.

Detailed interpretation: These results test the buy side of the fire-sale mechanism. Fire sales require not only forced sellers but also limited liquidity provision. When high-yield capital grows, the market can absorb downgraded bonds with smaller discounts.

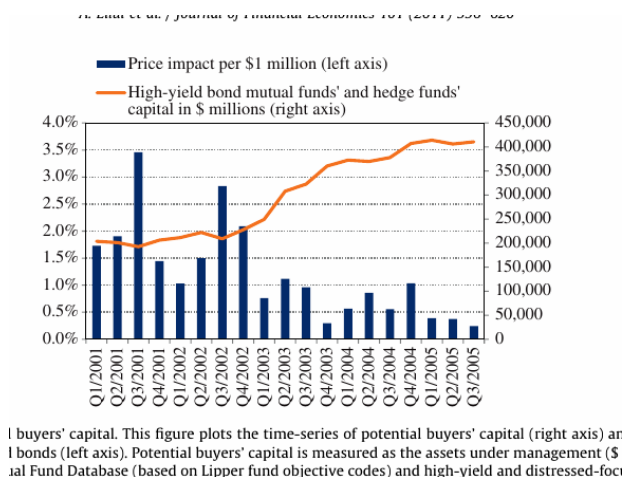
Economic message: Slow-moving capital matters: distressed sellers face larger discounts when specialized buyers have little capital.

Table 8
Price impact and potential buyers' capital.

This table relates price impact of insurance companies' selling of the downgraded bonds to potential buyers' capital. The downgrade is from investment to non-investment grades. Modified from Amihud (2002), price impact measure is calculated as the absolute value of price reversals divided by the total dollar trading volume of insurance companies (in millions) during the period that spans from event weeks -5 to $+5$. For each bond, the price reversal is measured as the log of the median trade price on or after event week $+21$ minus the log of the median trade price during event weeks -5 and $+5$. Potential buyers' capital is measured as the assets under management (\$ millions) of high-yield bond mutual funds from the CRSP Mutual Fund Database (based on Lipper fund objective codes) and high-yield and distressed-focused hedge funds from TASS (based on TASS classifications). The monthly potential buyers' capitals are averaged and reported by year. In the last four columns, the price impact is regressed on year dummies or directly on (demeaned) log of all potential buyers' capital, market movement controls, and (demeaned) bond characteristic controls. Market movement controls are measured over the same period of time as is the price impact and include stock market return, issuer's stock return, duration-adjusted return on 10-year Treasury note, duration-adjusted return on investment-grade corporate bond spread, duration-adjusted return on high-yield corporate bond spread, and the interactions of these variables with the lower than BB+ dummy. Bond characteristic controls include logged bond age at the time of downgrade, logged issue size, logged bond maturity at the time of downgrade, and the lower than BB+ dummy. Robust standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively.

Downgrade year	(1) High-yield bond mutual funds' capital	(2) High-yield bond mutual funds' and hedge funds' capital	(3) Price impact per \$1 million	(4) Price impact per \$1 million	(5) Price impact per \$1 million	(6) Price impact per \$1 million
2001	160,390	201,760	0.0188*** (0.0025)	0.0176*** (0.0026)	0.0131*** (0.0027)	
2002	163,530	217,489	0.0188*** (0.0020)	0.0133*** (0.0028)	0.0111*** (0.0029)	
2003	230,976	305,795	0.0089*** (0.0021)	0.0062** (0.0030)	0.0064** (0.0031)	
2004	260,684	385,994	0.0082*** (0.0014)	0.0083*** (0.0028)	0.0064** (0.0027)	
2005	254,512	411,685	0.0037*** (0.0006)	0.0032** (0.0014)	0.0009 (0.0019)	
ln(Total potential buyers' capital)						-0.0147*** (0.0046) 0.0079*** (0.0012)
Constant						
Controlled for market movement?			No	Yes	Yes	Yes
Controlled for bond characteristics?			No	Yes	Yes	Yes
Observations	55	55	495	404	404	404
Adjusted R-squared			0.096	0.105	0.160	0.185

(a) Table 8: Price impact and potential buyers' capital



(b) Figure 4: Price impact and potential buyers' capital

7.7 Figure 5: MCARs by Year of Downgrade + Table 9: Multivariate Regression of Price Reversals

Read:

- Figure 5 shows that bonds downgraded in 2001–2002 exhibit much larger negative MCARs and stronger reversals than bonds downgraded in 2003–2004.
- Table 9 shows that bond-level regulatory constraints predict larger price reversals.
- Coefficients remain positive and significant across alternative constraint measures, including average selling probability, average RBC ratio, and holdings by low-RBC insurers.
- Potential buyers' capital enters negatively, meaning more buyer capital reduces the magnitude of price reversals.

Detailed interpretation: Figure 5 and Table 9 combine the paper’s main time-series and multivariate evidence. The event-study pattern is not just a univariate artifact: after controlling for market movements, issuer stock returns, bond characteristics, and transaction costs, regulatory pressure still predicts fire-sale reversal.

Economic message: The strongest fire sales occur when constrained insurers own the bonds and outside capital is scarce.

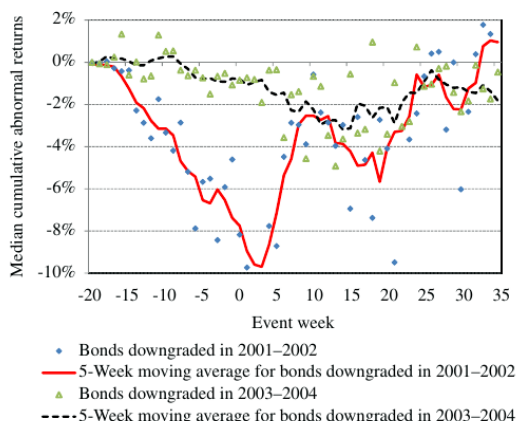


Figure 5: Median cumulative abnormal returns by year of downgrade. This figure plots the median cumulative abnormal returns (CARs) are calculated by the model:

(a) Figure 5: Median cumulative abnormal returns by year of downgrade

Table 9
Multivariate regression of price reversals.
This table reports coefficient estimates for the regression of price reversals on measures of regulatory constraint of an average insurance company holding the downgraded bonds. The downgrade is from investment to non-investment grades. The sample period is from 2001 to 2005. For each transaction executed during the period from event weeks -5 to +5, price reversal (the dependent variable) is measured as the logged benchmark price minus the logged transaction price. The benchmark price for each bond is measured as the median price of all transactions that are executed in or after event week -21. All other change and return variables are measured in the same fashion. Age and years to maturity are measured at the time of downgrade. Logged potential buyers' capital (\$ millions) is for the month of downgrade. Panel A reports OLS estimates for seven models with different measures of regulatory constraints, as listed in the column headings. White's robust standard errors are in parentheses. Panel B reports quantile regression estimates (0.25, 0.50, and 0.75) using two different measures of regulatory constraints, one using % holding by property insurance companies and the other using average holders' selling probability. Bootstrapped standard errors are in parentheses. *, **, and *** refer to statistical significance at 10%, 5%, and 1% levels, respectively. Variable definitions are in Appendix A and B.

	Bond-level regulatory constraint					
	% Holding by property insurers	Average selling probability	Average ln(RBC ratio)	% Holding by low RBC ratio	Average ln(RACR1)	% Holding by property insurers
Panel A: OLS regressions						
Bond-level regulatory constraint	0.0018*** (0.0004)	0.3502*** (0.0491)	-0.1208*** (0.0277)	0.0008*** (0.0169)	-0.1341*** (0.0222)	0.0011*** (0.0004)
ln(Buyers' capital)						0.3138*** (0.0467)
Insurance company controls						
Average ln(capital & surplus)	-0.0425*** (0.0041)	-0.0426*** (0.0041)	-0.0464*** (0.0042)	-0.0427*** (0.0042)		-0.0348*** (0.0040)
Average risky asset holding	0.0034*** (0.0009)	0.0027*** (0.0009)	0.0029*** (0.0009)	0.0029*** (0.0009)		-0.0005 (0.0008)
Market movement controls						
Stock market return	0.0429*** (0.0156)	0.0711*** (0.0157)	0.0631*** (0.0154)	0.0696*** (0.0154)	0.0678*** (0.0156)	0.1198*** (0.0146)
[Stock market return] × [Lower than BB + dummy]	-0.1097*** (0.0275)	-0.1232*** (0.0277)	-0.1100*** (0.0275)	-0.1122*** (0.0274)	-0.1238*** (0.0280)	-0.0927*** (0.0251)
Issuer's stock return	0.0706*** (0.0050)	0.0781*** (0.0050)	0.0746*** (0.0049)	0.0742*** (0.0049)	0.0709*** (0.0049)	0.0788*** (0.0047)
[Issuer's stock return] × [Lower than BB + dummy]	-0.0437*** (0.0096)	-0.0299*** (0.0096)	-0.0346*** (0.0097)	-0.0312*** (0.0097)	-0.0354*** (0.0092)	-0.0106 (0.0091)
Treasury bond return	0.3537*** (0.0609)	0.2924*** (0.0585)	0.2929*** (0.0571)	0.2895*** (0.0580)	0.2914*** (0.0575)	0.2469*** (0.0388)
[Treasury bond return] × [Lower than BB + dummy]	0.0453 (0.0874)	0.0463 (0.0866)	0.0350 (0.0854)	0.0450 (0.0860)	0.0265 (0.0860)	-0.1483*** (0.0709)
Investment-grade spread return	-0.1152 (0.1077)	-0.0900 (0.1036)	-0.0246 (0.1020)	-0.0628 (0.1018)	-0.0516 (0.1030)	-0.1437* (0.0797)
[Investment-grade spread return] × [Lower than BB + dummy]	0.2958* (0.1643)	0.2722* (0.1641)	0.2552 (0.1633)	0.2446 (0.1636)	0.2659 (0.1639)	0.3664** (0.1422)
High-yield spread return	0.1139*** (0.0252)	0.1063*** (0.0243)	0.0880*** (0.0240)	0.0962*** (0.0240)	0.0944*** (0.0240)	0.0415*** (0.0177)
[High-yield spread return] × [Lower than BB + dummy]	0.0233 (0.0404)	0.0391 (0.0403)	0.0270 (0.0399)	0.0385 (0.0402)	0.0239 (0.0400)	0.0400 (0.0358)
Transaction cost controls						
Δ Trade Direction	0.0144 (0.0111)	0.0173 (0.0107)	0.0180* (0.0108)	0.0196* (0.0107)	0.0151 (0.0107)	0.0358*** (0.0105)
Δ (Trade direction × ln(Trade size))	-0.0002 (0.0008)	-0.0002 (0.0008)	-0.0001 (0.0008)	-0.0003 (0.0008)	0.0000 (0.0008)	-0.0014* (0.0007)
Bond controls						
Lower than BB + dummy	0.0568*** (0.0045)	0.0556*** (0.0046)	0.0521*** (0.0046)	0.0501*** (0.0046)	0.0512*** (0.0045)	0.0510*** (0.0044)
ln(Age in years)	-0.0207*** (0.0030)	-0.0237*** (0.0036)	-0.0263*** (0.0037)	-0.0298*** (0.0034)	-0.0323*** (0.0033)	0.0174*** (0.0031)
ln(Issue size)	0.00018* (0.00020)	-0.0160*** (0.00024)	-0.0060*** (0.00022)	-0.0061*** (0.00021)	-0.0081*** (0.00021)	0.0299*** (0.00026)
ln(Years to maturity)	-0.0211*** (0.0033)	-0.0155*** (0.0033)	-0.0133*** (0.0033)	-0.0180*** (0.0033)	-0.0153*** (0.0033)	-0.0065*** (0.0031)
Constant	0.0303 (0.0292)	1.0243*** (0.0952)	1.1779*** (0.1025)	0.8912*** (0.0982)	1.1144*** (0.0965)	2.2835*** (0.0915)
Observations	4,067	4,067	4,067	4,067	4,067	4,067
Adjusted R-squared	0.405	0.428	0.425	0.426	0.427	0.488

(b) Table 9: Multivariate regression of price reversals, Panel A

7.8 Table 9 Continued + Table 10: Zero-Issuer-Stock-Return Sample

Read:

- Table 9 Panel B uses quantile regressions and confirms that regulatory pressure predicts reversals across the distribution.
- Table 10 focuses on bonds whose issuers have no significant stock return around the downgrade.
- This sample is designed to remove cases where the downgrade reveals major issuer-specific information.
- Even in this sample, more constrained bonds exhibit more negative MCARs and larger differences than less constrained bonds.

Detailed interpretation: Table 10 is one of the most important robustness blocks because it directly addresses the information-content critique. If issuer equity is not reacting, the downgrade is less likely to be major new information about firm value. The remaining differential price path is therefore more plausibly regulatory fire selling.

Economic message: The fire-sale pattern survives even when downgrade information is muted.

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	% Holding by property insurers			Average selling probability		
	25%	50%	75%	25%	50%	75%
Panel B: Quantile regressions						
Bond-level regulatory constraint	0.0012*** (0.0004)	0.0015*** (0.0004)	0.0005 (0.0007)	0.2496*** (0.0850)	0.4265*** (0.0638)	0.5037*** (0.0604)
ln(Buyers' capital)	-0.1553*** (0.0093)	-0.1894*** (0.0103)	-0.2479*** (0.0121)	-0.1639*** (0.0114)	-0.1800*** (0.0106)	-0.2352*** (0.0137)
Insurance company controls	Yes	Yes	Yes	Yes	Yes	Yes
Market movement controls	Yes	Yes	Yes	Yes	Yes	Yes
Transaction cost controls	Yes	Yes	Yes	Yes	Yes	Yes
Bond controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,067	4,067	4,067	4,067	4,067	4,067
Pseudo R-squared	0.247	0.301	0.387	0.261	0.323	0.404

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Cumulative abnormal returns for downgraded bonds with zero issuers' stock returns.

This table reports the median cumulative abnormal returns (MCARs) for the sample of downgraded bonds whose issuers' stock returns upon the downgrade are not significantly different from zero. Following Ambrose, Cai, and Helwege (2009), the zero-issuer-stock-return sample is identified by comparing the average daily stock return over the three-day event window (days -1 to +1) with the average historical stock return over 250 days (days -280 to -31). Using 5% significance level, the sample contains 584 bonds, but only 237 bonds meet the screening criteria and are used in the calculation of MCARs (Sample 63 in Table 1). The methods for (a) calculating abnormal returns (ARs), (b) accumulating ARs over event time to obtain cumulative abnormal returns (CARs) for each bond, and (c) aggregating CARs across bonds to obtain MCARs are as described in Table 6. The included bonds are divided into the more constrained and less constrained groups by (i) average selling probability (Panel A) calculated as described in Table 6 and (ii) years of downgrade (Panel B). Based on (i), the bonds with average selling probability above (below) the total sample median (Sample 4 in Table 1) are considered more (less) constrained. Based on (ii), the bonds downgraded in 2001/2002/2003/2004 are considered more (less) constrained. The t-statistics are for the tests of (a) null hypothesis that MCAR equals zero and (b) null hypothesis that MCARs for the more and less constrained groups are equal, and are calculated using bootstrapped standard errors (through median regressions of MCARs on a constant and a dummy for bonds in the more constrained group).

5-Week period	More constrained group		Less constrained group		Difference	
	(1) MCAR	HO: MCAR=0 t-statistic	(2) MCAR	HO: MCAR=0 t-statistic	(1)-(2) MCAR	HO: (1)=(2) t-statistic
Panel A: By average selling probability						
[-20, -16]	0.000	(0.00)	0.000	(0.00)	0.000	(0.00)
[-15, -11]	-1.262	(-2.52)	-0.389	(-1.79)	-0.874	(-1.88)
[-10, -6]	-2.442	(-1.67)	-0.351	(-1.00)	-2.090	(-1.18)
[-5, -1]	-3.445	(-4.23)	-0.833	(-2.21)	-2.613	(-2.65)
Week 0	-4.500	(-2.59)	-0.483	(-0.53)	-4.017	(-1.66)
[1, 5]	-3.960	(-2.66)	-0.162	(-0.50)	-3.798	(-2.66)
[6, 10]	-4.221	(-3.30)	-0.421	(-1.25)	-3.800	(-2.58)
[11, 15]	-2.567	(-2.16)	-1.204	(-1.98)	-1.363	(-1.03)
[16, 20]	-3.925	(-1.53)	-2.440	(-3.17)	-1.485	(-0.55)
[21, 25]	-1.815	(-1.08)	-1.064	(-1.04)	-0.752	(-0.61)
[26, 30]	-3.428	(-1.86)	0.275	(0.32)	-3.703	(-1.71)
[31, 35]	-0.523	(-0.27)	0.325	(0.38)	-0.849	(-0.42)
Panel B: By downgrade year						
[-20, -16]	0.019	(0.09)	0.000	(0.00)	0.019	(0.12)
[-15, -11]	-1.262	(-2.63)	-0.137	(-0.60)	-1.125	(-2.31)
[-10, -6]	-3.417	(-2.68)	0.359	(0.87)	-3.776	(-3.25)
[-5, -1]	-3.893	(-4.62)	-0.430	(-0.82)	-3.463	(-3.82)
Week 0	-5.637	(-1.09)	-0.091	(-0.08)	-5.546	(-1.21)
[1, 5]	-3.881	(-2.59)	-0.063	(-0.14)	-3.817	(-2.58)
[6, 10]	-2.343	(-1.00)	-1.007	(-1.64)	-1.336	(-0.42)
[11, 15]	-3.149	(-3.41)	-1.145	(-2.09)	-2.004	(-1.95)
[16, 20]	-4.218	(-3.44)	-1.882	(-1.32)	-2.336	(-1.25)
[21, 25]	-3.692	(-1.67)	-0.763	(-0.77)	-2.928	(-1.12)
[26, 30]	-1.282	(-0.94)	-0.080	(-0.12)	-1.202	(-0.81)
[31, 35]	0.346	(0.24)	-0.885	(-1.11)	1.231	(0.70)

holds whether we use average selling probability or years of Helwege (2009) in that they find generally limited price

(a) Table 9: Multivariate regression of price reversals, Panel B (b) Table 10: Cumulative abnormal returns for zero-issuer-stock-return bonds

7.9 Table 11: Cumulative Abnormal Returns by Alternative Methodologies

Read:

- Table 11 recalculates MCARs using bond-index adjustment and mean adjustment methods.
- The more constrained group still has more negative MCARs than the less constrained group.
- Results hold when groups are defined by average selling probability and by downgrade year.
- The robustness checks show that the main result is not driven by one particular abnormal-return model.

Detailed interpretation: This table addresses methodology. Corporate bond abnormal returns are difficult because trades are infrequent and transaction costs are large. The paper shows that the regulatory-pressure pattern remains under alternative approaches.

Economic message: The fire-sale conclusion is robust to the way abnormal returns are calculated.

Table 11

Cumulative abnormal returns by alternative methodologies.

This table reports the median cumulative abnormal returns (MCARs), calculated by two alternative methodologies of measuring abnormal returns: (i) bond-index adjustment (Panels A and C), and (ii) mean adjustment (Panels B and D). *Bond-index adjusted* abnormal returns are calculated as actual bond returns minus rating- and maturity group-matched Barclays bond (price) index return. *Mean adjusted* abnormal returns are calculated as actual bond excess returns minus the mean excess return before event week -20 (scaled by number of days between trades). Excess returns are defined as the difference between corporate bond returns and returns on Treasury bonds in the same maturity group. The calculation uses only the zero-issuer-stock-return sample, Sample 6.1 in Table 1. The methods for (a) accumulating abnormal returns over event time to obtain cumulative abnormal returns (CARs) for each bond, and (b) aggregating CARs across bonds to obtain MCARs are as described in Table 6. The included bonds are divided into the more constrained and less constrained groups by (i) average selling probability (Panels A and B) calculated as described in Table 6 and (ii) years of downgrade (Panels C and D). Based on (i), the bonds with average selling probability above (below) the total sample median (Sample 4 in Table 1) are considered more (less) constrained. Based on (ii), the bonds downgraded in 2001/2002 (2003/2004) are considered more (less) constrained. The t -statistics are for the tests of (a) null hypothesis that MCAR equals zero and (b) null hypothesis that MCARs for the more and less constrained groups are equal, and are calculated using bootstrapped standard errors (through median regressions of MCARs on a constant and a dummy for bonds in the more constrained group).

5-Week period	More constrained group		Less constrained group		Difference	
	(1) MCAR	H0: MCAR=0 t -statistic	(2) MCAR	H0: MCAR=0 t -statistic	(1) - (2) MCAR	H0: (1)=(2) t -statistic
<i>Panel A: Index-adjusted MCARs by average selling probability</i>						
[−10, −6]	−3.260	(−3.70)	−0.039	(−0.08)	−3.221	(−2.73)
[−5, −1]	−4.134	(−3.96)	−0.271	(−0.37)	−3.862	(−3.09)
Week 0	−3.762	(−1.88)	−0.767	(−1.25)	−2.995	(−1.62)
[1, 5]	−3.979	(−2.93)	−0.876	(−1.49)	−3.103	(−2.07)
[6, 10]	−3.307	(−2.84)	−1.975	(−3.20)	−1.331	(−1.12)
<i>Panel B: Mean-adjusted MCARs by average selling probability</i>						
[−10, −6]	−3.015	(−2.95)	−1.196	(−1.62)	−1.818	(−1.36)
[−5, −1]	−4.868	(−5.09)	−2.332	(−2.74)	−2.536	(−1.87)
Week 0	−4.291	(−2.16)	−2.113	(−3.66)	−1.978	(−0.82)
[1, 5]	−5.205	(−4.71)	−1.583	(−2.47)	−3.622	(−2.93)
[6, 10]	−5.289	(−2.99)	−1.256	(−2.04)	−4.033	(−1.89)
<i>Panel C: Index-adjusted MCARs by downgrade year</i>						
[−10, −6]	−5.720	(−5.60)	−0.285	(−0.53)	−5.435	(−3.85)
[−5, −1]	−5.398	(−2.94)	−0.394	(−0.97)	−5.004	(−2.83)
Week 0	−4.426	(−1.77)	−0.850	(−0.82)	−3.576	(−1.02)
[1, 5]	−5.724	(−3.14)	−0.876	(−1.85)	−4.848	(−2.38)
[6, 10]	−3.784	(−2.15)	−2.434	(−3.89)	−1.350	(−0.73)
<i>Panel D: Mean-adjusted MCARs by downgrade year</i>						
[−10, −6]	−6.938	(−5.15)	0.658	(1.46)	−7.596	(−4.86)
[−5, −1]	−7.168	(−4.17)	0.584	(0.66)	−7.752	(−4.10)
Week 0	−4.927	(−2.11)	0.875	(0.93)	−5.801	(−3.21)
[1, 5]	−5.695	(−6.00)	−0.150	(−0.28)	−5.545	(−4.45)
[6, 10]	−3.382	(−1.82)	−0.658	(−1.51)	−2.724	(−1.39)

Table 11: Cumulative abnormal returns by alternative methodologies

8 Detailed result map

- **Mechanism result.** Low-capital insurers are more likely to sell fallen angels, as shown in Table 4.
- **Volume result.** Net selling spikes around the downgrade, especially for low-RBC insurers, as shown in Table 5 and Figure 1.
- **Price result.** High regulatory-pressure bonds experience large temporary MCAR declines and subsequent recovery, as shown in Figure 2 and Table 6.
- **Sell-side state dependence.** Insurance industry distress increases forced selling, as shown in Table 7 and Figure 3.
- **Buy-side state dependence.** Scarce high-yield and distressed-investor capital makes price impacts larger, as shown in Table 8 and Figure 4.
- **Information robustness.** Zero-stock-return issuer samples and alternative abnormal-return methodologies preserve the main pattern, as shown in Tables 10 and 11.
- **Main interpretation.** Regulatory constraints generate coordinated sales in an illiquid market, and prices overshoot downward until outside capital arrives.

9 Short summary

- The paper studies corporate bonds downgraded from investment grade to speculative grade during 2001–2005.
- Insurance companies are central because they hold large quantities of investment-grade bonds and face stricter capital requirements for speculative-grade bonds.
- The main source of variation is cross-sectional variation in insurer regulatory constraints and bond-level exposure to constrained insurers.

- Constrained insurers are significantly more likely to sell downgraded bonds immediately after the downgrade.
- Bonds with higher predicted forced selling have larger price drops and subsequent reversals.
- The price discount is temporary and largely disappears by roughly 30–35 weeks after the event.
- Fire-sale effects are larger when the insurance industry is distressed and when outside buyer capital is scarce.
- The paper uses issuer stock returns, zero-stock-return samples, and alternative abnormal-return methods to address information explanations.
- The central lesson is that regulation can generate synchronized selling and temporary mispricing when regulated institutions hold similar assets.

10 Conclusion

10.1 Core conclusion

Ellul, Jotikasthira, and Lundblad show that regulatory constraints imposed on insurance companies can create fire sales in the corporate bond market. When a bond is downgraded from investment grade to speculative grade, constrained insurers become more likely to sell. Bonds held by these constrained insurers then experience large temporary price declines and subsequent reversals.

10.2 Economic intuition

The intuition is a supply-demand imbalance. A downgrade moves a bond into a higher regulatory-risk category. Insurers with weak capital positions have strong reasons to reduce exposure. But other insurers face the same regulation and may be unable or unwilling to buy. Therefore liquidity must come from outside investors, whose capital is limited and slow-moving. Prices fall to compensate buyers for providing liquidity and bearing inventory risk. As more buyers enter, prices recover.

10.3 Incremental contribution

The paper extends the fire-sale literature by identifying regulation as a source of forced selling. It also demonstrates the value of micro-level institutional data: without observing insurer-specific financial health and actual trades, it would be difficult to distinguish forced selling from generic downgrade information. The paper’s incremental insight is that the same credit event can have different price effects depending on who holds the bond before the event.

10.4 Policy implication

Capital regulation can improve solvency but can also have market-level liquidity consequences. If many institutions face the same rule and hold the same assets, a common shock can force synchronized sales. This creates a tension between microprudential safety and market-wide liquidity.

10.5 Methodological implication

The paper illustrates how to study fire sales when fundamental value is unobservable. The relevant test is not simply whether prices fall, but whether they fall more for assets held by constrained sellers and then reverse when forced selling pressure fades.

10.6 Limitations

The paper cannot directly observe every insurer's internal regulatory decision, nor can it observe the full set of potential buyers. Corporate bonds trade irregularly, which limits event-study precision. Downgrades may also contain information that is difficult to remove completely.

10.7 Takeaway

Regulation matters for asset prices when regulated institutions are large holders of illiquid assets. Downgrades can trigger fire-sale dynamics not because all bonds become equally worse, but because some bonds are held by institutions that are forced to sell at the wrong time.